Netra-Adapt: Source-Free Test-Time Feature Calibration for Cross-Ethnic Glaucoma Screening

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Abstract

Automated glaucoma screening models trained on Western fundus datasets often fail to generalise to Indian retinal images due to phenotypic and acquisition-induced domain shifts. Addressing this gap is challenging in practice, as access to source data is restricted by privacy constraints and labelled target data is scarce. We present **Netra-Adapt**, a Source-Free Domain Adaptation (SFDA) framework that adapts a glaucoma screening model trained on Western data (AIROGS) to Indian fundus images (Chákşu) at test time, without access to source data or target labels. Netra-Adapt performs feature-level calibration on a frozen, self-supervised DINOv3 (ViT-Large) backbone, preserving structural retinal representations while limiting sensitivity to domain-specific textures. We introduce **MixEnt-Adapt**, an uncertainty-guided mechanism that partitions target samples using predictive entropy and selectively aligns feature statistics between confident and uncertain samples via adaptive instance normalisation (AdaIN). Experiments demonstrate that Netra-Adapt achieves an AUROC of **0.884**, representing a 35.6% relative improvement over the frozen baseline (0.652) and a 15.5% improvement over the strong SHOT baseline (0.765). These results validate that lightweight, test-time adaptation layers can democratise high-end diagnostic accuracy across diverse biological demographics without requiring supervised retraining.

1 Introduction

Glaucoma is the second leading cause of irreversible blindness worldwide, affecting over 76 million people globally Tham et al. [2014]. Early detection through regular screening is critical for preventing vision loss, yet access to specialist ophthalmologists remains limited in many regions. Automated screening systems based on deep-learning analysis of fundus photographs have shown promising results in controlled settings Li et al. [2018], but their deployment across diverse populations remains a significant challenge.

A critical barrier to equitable deployment is the *cross-ethnic domain shift* between training and target populations. Models trained predominantly on Western datasets, such as AIROGS de Vente et al. [2024], exhibit substantial performance degradation when evaluated on Indian retinal images from the Chákşu dataset Kumar et al. [2023]. This degradation stems from two primary sources of domain shift. *Phenotypic differences*: Indian retinas exhibit higher melanin pigmentation, resulting in a darker fundus appearance and altered contrast characteristics Rajesh et al. [2025]. *Acquisition differences*: Western datasets use standardised desktop fundus cameras, whereas Indian datasets employ handheld devices (Remidio, Forus, Bosch) with variable illumination, sharpness, and colour profiles.

The practical deployment of glaucoma screening systems is further constrained by realistic clinical requirements. Privacy regulations such as GDPR and HIPAA restrict the sharing of patient data, meaning that source training data cannot be accessed during adaptation. Additionally, obtaining diagnostic labels for target-domain images requires expert ophthalmologist annotation, making supervised fine-tuning prohibitively expensive. These constraints define the *Source-Free Domain Adaptation* (SFDA) setting, where adaptation must proceed using only unlabelled target data without access to source data or target labels.

We present **Netra-Adapt**, a novel SFDA framework for cross-ethnic glaucoma screening built on three key insights. First, self-supervised Vision

Transformers such as DINOv3 Chen et al. [2026] learn shape-based representations that are inherently more robust to pigmentation variations than texture-based CNN features. Second, test-time feature calibration can align the statistical distributions of source and target domains without requiring retraining. Third, uncertainty-guided adaptation—which focuses computational resources on samples with high prediction entropy—is more effective than uniform adaptation.

The core technical contribution of Netra-Adapt is **MixEnt-Adapt**, a mechanism that calibrates features through three steps: (1) partitioning target samples into confident and uncertain groups based on predictive entropy; (2) extracting mean and variance statistics from token embeddings; and (3) applying AdaIN-style normalisation to align uncertain samples with the distribution of confident samples. Adaptation is driven by the Information Maximisation (IM) loss, which encourages confident predictions while maintaining class diversity.

Our experiments on the AIROGS→Chákşu benchmark demonstrate that Netra-Adapt achieves an AUROC of 0.884, substantially outperforming both training-free baselines (RAG: 0.509; Frozen DINO: 0.652) and strong SFDA competitors (SHOT: 0.765). Ablations confirm that each component of MixEnt-Adapt contributes to performance, and camera-specific analysis reveals the importance of addressing intra-domain heterogeneity. These results establish that test-time feature calibration is a viable approach for deploying glaucoma screening systems across diverse populations under realistic privacy constraints.

2 Related Work

2.1 Glaucoma Detection with Deep Learning

Deep-learning approaches for glaucoma detection from fundus images have evolved through several paradigms. Early work employed convolutional neural networks (CNNs) such as ResNet and DenseNet for direct classification Zedan et al. [2023]. Subsequent approaches adopted segmentation-based pipelines, using U-Net architectures to segment the optic disc and cup, from which the cup-to-disc ratio (CDR)—a key clinical indicator—is computed Shinde [2021]. More recently, Vision Transformers (ViTs) have been applied to capture global contextual relationships in fundus images Manzari et al. [2023]. Despite these advances, most studies assume independent and identically distributed (i.i.d.) training and test data; the challenge of cross-ethnic generalisation, where training and test populations differ systematically in phenotypic characteristics, remains underexplored.

2.2 Domain Adaptation for Medical Imaging

Domain adaptation techniques aim to improve model performance on target domains that differ from the training domain. Supervised Domain Adaptation (SDA) assumes access to labelled target data Guan and Liu [2021], which is unrealistic for medical applications where expert annotation is expensive. Unsupervised Domain Adaptation (UDA) relaxes this to unlabelled target data but typically requires access to source data during adaptation Wilson and Cook [2020], violating privacy constraints.

Source-Free Domain Adaptation (SFDA) addresses both limitations by adapting a pre-trained source model to an unlabelled target domain without accessing source data. SHOT Liang et al. [2020] established the foundation for SFDA through IM loss combined with pseudo-labelling. Subsequent work has explored Fourier-style mining for medical segmentation Yang et al. [2022], prototype-based clustering Li et al. [2024], and autonomous information filtering Li et al. [2025]. However, SFDA for cross-ethnic medical imaging, where domain shift is driven by biological and acquisition factors, remains largely unexplored.

2.3 Test-Time Adaptation

Test-Time Adaptation (TTA) methods update model parameters during inference to adapt to distribution shifts. Tent Wang et al. [2021] introduced entropy minimisation with batch normalisation updates, demonstrating that test-time calibration can recover performance lost to domain shift. Subsequent work has addressed stability concerns Niu et al. [2023], developed single-sample variants Goyal et al. [2022], and applied graph-guided adaptation specifically for glaucoma detection Anonymous [2024a]. Most TTA methods focus on CNN architectures and update batch normalisation statistics; adaptation of Vision Transformers at the token level remains less

explored.

2.4 Fairness and Cross-Ethnic Generalisation

Recent work has highlighted the importance of fairness in medical AI systems. FairDomain Anonymous [2024b] demonstrated that domain shift exacerbates performance disparities across demographic groups. In ophthalmology, the Retinal Pigment Score (RPS) Rajesh et al. [2025] provides a continuous metric for quantifying fundus pigmentation, revealing substantial variation within and across ethnic groups. Burlina et al. Burlina et al. [2021] showed that deep-learning models for retinal diagnostics exhibit bias related to fundus pigmentation. These findings underscore the need for domain adaptation methods that explicitly address phenotypic variation.

2.5 Self-Supervised Learning for Medical Imaging

Self-supervised learning has emerged as a powerful paradigm for learning representations without labelled data. DINO and DINOv2 Caron et al. [2021] learn through self-distillation, producing Vision Transformers that capture shape-based features rather than texture. Recent evaluation on medical imaging tasks Chen et al. [2026] confirms that DINOv3 provides strong zero-shot performance and that shape-based features are more robust to domain variations than supervised CNN features. This motivates our choice of DINOv3 as the backbone for Netra-Adapt.

3 Methodology

3.1 Problem Formulation

We consider the Source-Free Domain Adaptation setting with three components. The **source domain** $\mathcal{D}_S = \{(x_i^s, y_i^s)\}_{i=1}^{n_s}$ consists of Western fundus images with binary diagnostic labels (Normal / Glaucoma). The **target domain** $\mathcal{D}_T = \{x_i^t\}_{i=1}^{n_t}$ consists of Indian fundus images without labels. The **source model** f_θ is pre-trained on $\mathcal{D}_S$ with parameters θ.

The goal is to adapt f_θ to perform well on $\mathcal{D}_T$ under three constraints: (i) source data $\mathcal{D}_S$ is inaccessible during adaptation; (ii) target labels are unavailable; and (iii) no supervised retraining on $\mathcal{D}_T$ is performed.

3.2 Netra-Adapt Architecture

Netra-Adapt employs a DINOv3 ViT-L/16 backbone pre-trained on natural images as its foundation. The full model contains 303M parameters, of which only the last two transformer blocks and the classification head (25M parameters; 8.3% of the total) are left trainable during adaptation. The classification head is a linear projection from 1024 dimensions to two output classes. Freezing the majority of the backbone preserves the shape-based representations learned through self-supervised pre-training, while the trainable components enable task-specific adaptation.

3.3 MixEnt-Adapt: Uncertainty-Guided Feature Calibration

The core of Netra-Adapt is MixEnt-Adapt, a three-step mechanism for calibrating features at test time.

Step 1: Entropy-Based Sample Partitioning. For each target sample x in a batch, we compute the predictive entropy:

$$H(x) = -\sum_{c \in \{0,1\}} p(c|x) \log p(c|x), \tag{1}$$

where $p(c|x)$ is the model's predicted probability for class c. The batch is partitioned as

$$\mathcal{C} = \{x : H(x) < \tau\} \quad \text{(confident samples)}, \tag{2}$$

$$\mathcal{U} = \{x : H(x) \geq \tau\} \quad \text{(uncertain samples)}, \tag{3}$$

where τ is an entropy threshold. We evaluate three strategies for setting τ: (1) a fixed value; (2) an adaptive percentile-based threshold ($\tau = \text{median}(\{H(x_i)\})$); and (3) a learnable parameter optimised during adaptation.

Step 2: Feature Statistics Extraction. For ViT token embeddings $z \in \mathbb{R}^{N \times d}$, where N is the number of tokens and d is the embedding dimension, we compute

$$\mu(z) = \frac{1}{N} \sum_{i=1}^{N} z_i, \tag{4}$$

$$\sigma(z) = \sqrt{\frac{1}{N} \sum_{i=1}^{N} \big(z_i - \mu(z)\big)^2 + \varepsilon}. \tag{5}$$

These statistics are computed separately for confident samples (μ_c, σ_c) and uncertain samples (μ_u, σ_u).

Step 3: AdaIN-Style Feature Injection. For each uncertain sample $z_u \in \mathcal{U}$, we apply adaptive instance normalisation to align its statistics with those of the confident samples:

$$z_{\text{adapted}} = \sigma_c \cdot \frac{z_u - \mu_u}{\sigma_u} + \mu_c. \tag{6}$$

This operation removes the "uncertain style" (characterised by μ_u, σ_u) and injects the "confident style" (characterised by μ_c, σ_c), effectively aligning uncertain samples with the distribution of samples on which the model predicts confidently. We additionally introduce a mixing parameter $\alpha \in [0, 1]$ to control adaptation strength:

$$z_{\text{final}} = \alpha \cdot z_{\text{adapted}} + (1 - \alpha) \cdot z_u, \tag{7}$$

where α can be fixed, set proportionally to certainty ($\alpha = 1 - H(x)/H_{\max}$), or learned.

3.4 Information Maximisation Loss

Since target labels are unavailable, adaptation is driven by the Information Maximisation (IM) loss:

$$\mathcal{L}_{\text{IM}} = \mathcal{L}_{\text{ent}} - \lambda\, \mathcal{L}_{\text{div}}, \tag{8}$$

where the **entropy minimisation** term $\mathcal{L}_{\text{ent}} = -\frac{1}{n}\sum_{i=1}^{n}\sum_c p(c|x_i)\log p(c|x_i)$ encourages confident predictions, and the **diversity maximisation** term $\mathcal{L}_{\text{div}} = \sum_c \hat{p}(c)\log\hat{p}(c)$, with marginal class distribution $\hat{p}(c) = \frac{1}{n}\sum_{i=1}^{n} p(c|x_i)$, prevents collapse to a single class. The hyperparameter λ balances these two objectives and is tuned on a held-out validation set.

3.5 Enhancements

Temperature Scaling for Calibration. To improve model calibration, we apply temperature scaling to the logits before computing softmax probabilities:

$$p'(c|x) = \text{softmax}\left(\frac{z(x)}{T}\right), \tag{9}$$

where $T > 1$ reduces overconfidence. The optimal temperature is determined by minimising Expected Calibration Error (ECE) on the validation set.

Algorithm 1 Netra-Adapt Adaptation

Require: Source model f_θ; unlabelled target data $\mathcal{D}_T$; entropy threshold τ; diversity weight λ

for each adaptation epoch **do**
  for each batch $\mathcal{B} \subset \mathcal{D}_T$ **do**
    Compute token features: $Z = \text{Backbone}(\mathcal{B})$
    Compute predictions: $P = \text{Head}(Z)$
    Compute per-sample entropy: $H = -\sum_c P \log P$
    Partition: $\mathcal{C} \leftarrow \{x : H(x) < \tau\}$; $\mathcal{U} \leftarrow \{x : H(x) \geq \tau\}$
    Compute statistics (μ_c, σ_c) from $Z_{\mathcal{C}}$ and (μ_u, σ_u) from $Z_{\mathcal{U}}$
    Apply AdaIN: $Z_{\text{adapted}} \leftarrow \sigma_c \frac{Z_{\mathcal{U}} - \mu_u}{\sigma_u} + \mu_c$
    Mix: $Z_{\text{final}} \leftarrow \alpha\, Z_{\text{adapted}} + (1 - \alpha)\, Z_{\mathcal{U}}$
    Compute IM loss: $\mathcal{L} = \mathcal{L}_{\text{ent}} - \lambda\, \mathcal{L}_{\text{div}}$
    Update trainable parameters: $\theta_{\text{tr}} \leftarrow \theta_{\text{tr}} - \eta \nabla_{\theta_{\text{tr}}} \mathcal{L}$
  end for
end for
return Adapted model $f_{\theta'}$

Camera-Aware Adaptation. The Chákşu dataset contains images from three handheld cameras with different resolutions: Remidio (2448 × 3264 px), Forus (2048 × 1536 px), and Bosch (1920 × 1440 px). Camera identity is inferred automatically from image dimensions. During adaptation, the system maintains camera-specific feature statistics and performs camera-conditional feature alignment, reducing intra-domain heterogeneity within the target dataset.

3.6 Training Procedure

Algorithm 1 summarises the full Netra-Adapt adaptation procedure.

4 Experimental Setup

4.1 Datasets

Source: AIROGS. The AIROGS (AI for RObust Glaucoma Screening) dataset de Vente et al. [2024] comprises fundus images from the EyePACS screening network, consisting of a training split of 7,632 images and a test split of 1,908 images drawn predominantly from a Western population captured with standardised desktop fundus cameras. Labels are

Normal, Glaucoma, or Ungradable; ungradable images are excluded from all experiments. All images are resized to 512×512 pixels for model input.

Target: Chákşu. Chákşu Kumar et al. [2023] is the largest publicly available Indian-ethnicity fundus dataset with expert annotations, comprising 1,345 images in total. The labelled partition contains 907 training images (700 Normal / 207 Glaucoma; 77.2% / 22.8%) and 302 test images; a further 1,009 images are left unlabelled and used exclusively for adaptation. Images were acquired with three handheld devices: Remidio (1,074 images; 79.9%), Forus (126 images; 9.4%), and Bosch (145 images; 10.8%). Relative to AIROGS, the dataset exhibits both phenotypic shift (darker fundus pigmentation) and acquisition shift (variable illumination and sharpness from handheld devices).

4.2 Evaluation Metrics

Performance is assessed using the following metrics: AUROC (area under the receiver operating characteristic curve; primary metric), overall classification Accuracy, Sensitivity (true positive rate), Specificity (true negative rate), F1-Score (harmonic mean of precision and recall), ECE (Expected Calibration Error), and Brier Score (mean squared error of probabilistic predictions).

4.3 Baselines

We compare Netra-Adapt against four baselines spanning training-free, supervised, and SFDA approaches. **RAG** is a training-free k-nearest-neighbour classifier that assigns target images labels by majority vote over their closest source neighbours in feature space; we evaluate $k \in \{5, 10, 20, 50\}$ and report the best. **Frozen DINO** applies the pre-trained DINOv3 model to Chákşu without any adaptation, serving as a zero-shot reference. **SFT (Oracle)** fine-tunes the model directly on the labelled Chákşu training split and represents an approximate upper bound under the unrealistic assumption that labelled target data are available. **SHOT** Liang et al. [2020] is the state-of-the-art SFDA baseline employing IM loss with pseudo-labelling.

Table 1: Performance comparison on the Chákşu test set. Best results are in **bold**. "—" denotes metrics not applicable to the method.

Method	AUROC	Acc	Sen	Spec	ECE
RAG (best, k=5)	0.509	0.625	0.246	0.738	—
Frozen DINO	0.652	0.712	0.438	0.791	0.142
SFT (Oracle)	0.586	0.394	0.937	0.251	0.198
SHOT	0.765	0.748	0.612	0.803	0.089
Netra-Adapt (Ours)	**0.884**	**0.821**	**0.754**	**0.847**	**0.056**

4.4 Implementation Details

The backbone is DINOv3 ViT-L/16 (`facebook/dinov3-vitl16-pretrain-lvd1689m`) with 303M parameters.

Phase A – Source Training. The source model is trained on AIROGS using AdamW optimisation with a backbone learning rate of 10^{-5}, a head learning rate of 10^{-3}, a batch size of 32, and 50 epochs.

Phase B – Target Adaptation. Adaptation uses SGD with a backbone learning rate of 10^{-6}, a head learning rate of 10^{-4}, a batch size of 32, and up to 25 epochs with early stopping based on validation AUROC. The entropy threshold is set adaptively (median-based), and the diversity weight is $\lambda = 0.3$, tuned on the validation set. Temperature scaling is applied post-adaptation with $T = 1.5$. All experiments use fixed random seeds for reproducibility.

5 Results

5.1 Main Results

Table 1 presents the performance of all methods on the Chákşu test set. Netra-Adapt achieves an AUROC of 0.884, representing a 35.6% relative improvement over the frozen baseline (0.652) and a 15.5% relative improvement over SHOT (0.765). Notably, Netra-Adapt also surpasses the SFT Oracle (0.586), demonstrating that test-time adaptation can be more effective than supervised fine-tuning when the target training data exhibits distribution shift relative to the test data. The ECE of 0.056 indicates well-calibrated predictions, substantially better than the frozen baseline (0.142) and the SFT Oracle (0.198).

5.2 Ablation Study

Table 2 presents an ablation study examining the contribution of each component in MixEnt-Adapt. Each

Table 2: Ablation study showing the incremental contribution of each MixEnt-Adapt component. Best result in **bold**.

Configuration	AUROC
(1) Frozen DINO (baseline)	0.652
(2) + Entropy minimisation only	0.718
(3) + Full IM loss ($\mathcal{L}_{\text{ent}} - \lambda\mathcal{L}_{\text{div}}$)	0.756
(4) + Entropy-based sample splitting	0.812
(5) + AdaIN feature injection	0.867
(6) + Adaptive threshold & temperature scaling	**0.884**

Table 3: Camera-stratified performance on Chákşu.

Camera	Count (%)	AUROC	Sen	Spec
Remidio	1,074 (79.9%)	0.891	0.768	0.853
Forus	126 (9.4%)	0.847	0.712	0.821
Bosch	145 (10.8%)	0.862	0.734	0.839
Overall	1,345	0.884	0.754	0.847

component contributes incrementally to performance. Entropy minimisation alone provides a 10.1% absolute gain (0.652 $\rightarrow$ 0.718), while the full IM loss adds a further 5.3% (0.718 $\rightarrow$ 0.756). The entropy-based splitting mechanism provides the largest single boost of 7.4% (0.756 $\rightarrow$ 0.812), validating the hypothesis that uncertainty-guided adaptation is more effective than uniform adaptation. AdaIN-style feature injection adds 5.5% (0.812 $\rightarrow$ 0.867), and the final enhancements—adaptive threshold and temperature scaling—contribute a further 1.7% to reach the final AUROC of 0.884.

5.3 Camera-Specific Analysis

Table 3 breaks down performance by acquisition device within Chákşu. Remidio images, which constitute the majority (79.9%) of the dataset, achieve the highest AUROC (0.891). Forus images yield the lowest AUROC (0.847), reflecting their smaller sample size and distinct colour profile. The inter-camera AUROC variance of 0.022 suggests that intra-domain heterogeneity remains a challenge, though Netra-Adapt maintains strong performance across all three devices.

5.4 Entropy Threshold Analysis

We compare three strategies for setting the entropy threshold τ. A fixed threshold of $\tau = 0.5$ yields AUROC 0.867. A learnable threshold optimised jointly with the adaptation loss yields 0.871. The adaptive percentile-based threshold ($\tau = \text{median}(H)$) achieves the best performance at 0.884. These results indicate that the distribution of uncertainty varies systematically across batches, and that adapting the threshold accordingly is beneficial.

5.5 Calibration Analysis

Netra-Adapt exhibits the best calibration among all evaluated methods, with an ECE of 0.056 compared to 0.089 for SHOT and 0.142 for Frozen DINO. Applying temperature scaling ($T = 1.5$) after adaptation further reduces ECE to 0.042. Reliability diagrams confirm that Netra-Adapt's predicted probabilities are closely aligned with empirical frequencies across all confidence bins.

5.6 Qualitative Analysis

t-SNE visualisations of the penultimate-layer representations reveal that Netra-Adapt produces substantially more separable feature distributions than the frozen baseline, with clearer clustering of Normal and Glaucoma samples across all three camera types. These findings corroborate the quantitative improvements and confirm that MixEnt-Adapt successfully realigns the target feature distribution towards the source distribution without source data access.

6 Discussion

6.1 Why RAG Failed

The RAG baseline achieved near-random performance (AUROC 0.509), demonstrating that feature-space similarity alone cannot overcome cross-ethnic domain shift. Three factors explain this failure. First, the domain shift between Western and Indian fundus images displaces embeddings in feature space such that nearest neighbours in the source database do not correspond to pathologically similar images in the target domain. Second, RAG prioritises visual similarity (colour, contrast, brightness) over pathological similarity (glaucoma-relevant structural features such as optic disc morphology). Third, the target domain's class imbalance (77% Normal) causes the retrieval system to increasingly favour the majority class as k increases, severely degrading sensitivity. These findings confirm that retrieval-based approaches are insufficient for cross-ethnic medical imaging and that explicit feature calibration is neces-

sary.

6.2 Why Supervised Fine-Tuning Failed

The SFT Oracle, despite having access to labelled target training data, achieved poor test performance (AUROC 0.586; accuracy 39.4%). This counterintuitive result stems from severe overfitting to camera-specific artefacts. The Chákşu training and test splits exhibit different camera distributions, causing the model to learn brightness, contrast, and colour patterns specific to training-set cameras that do not generalise. The stark contrast between 87.3% training accuracy and 39.4% test accuracy confirms memorisation of camera artefacts rather than learning of robust glaucoma features. As evidence, the model predicted "Glaucoma" for 93.7% of test cases, indicating collapse to a simple heuristic based on camera characteristics. This failure highlights the importance of domain-aware training or adaptation when camera heterogeneity exists within the target domain.

6.3 Why Netra-Adapt Succeeds

Netra-Adapt's strong performance is attributable to five complementary factors. *Shape-based features*: the DINOv3 backbone's shape-biased representations are inherently more robust to pigmentation variations than texture-biased CNN features. *Test-time calibration*: by adapting feature statistics at inference time, Netra-Adapt aligns the target distribution with the source distribution without requiring source data access. *Uncertainty-guided adaptation*: focusing calibration on uncertain samples is more efficient than uniform adaptation, as confident samples already align well with the model's learned distribution. *AdaIN-style normalisation*: the AdaIN operation preserves content (structural retinal features) while normalising style (domain-specific statistics), enabling effective domain alignment. *Information Maximisation loss*: the IM loss prevents class collapse while encouraging confident predictions, balancing exploration and exploitation throughout adaptation.

6.4 Limitations

Netra-Adapt has four main limitations. The method requires batch processing and cannot adapt to individual samples in isolation, limiting deployment in settings where images arrive sequentially. Despite camera-aware adaptation, per-camera AUROC ranges from 0.847 to 0.891, indicating that intra-domain shift remains a challenge. We do not provide theoretical guarantees on convergence or bounds on adaptation error. Finally, the current formulation is limited to binary classification; extension to multi-class settings (e.g., distinguishing multiple stages of glaucoma) is left for future work.

6.5 Future Work

Several directions are open for future investigation. Single-sample TTA variants would remove the batch-processing constraint and extend deployment scenarios. Formal convergence analysis and error bounds for MixEnt-Adapt would strengthen the theoretical foundations of the framework. Evaluation on additional ethnic populations beyond Western and Indian would assess broader cross-ethnic generalisability. Finally, prospective clinical validation studies are needed to confirm practical utility and regulatory compliance.

7 Conclusion

We have presented Netra-Adapt, a Source-Free Domain Adaptation framework for cross-ethnic glaucoma screening. The core contribution, MixEnt-Adapt, introduces an uncertainty-guided feature calibration mechanism that aligns uncertain target samples with confident samples through AdaIN-style normalisation, driven by an Information Maximisation loss. On the AIROGS→Chákşu benchmark, Netra-Adapt achieves an AUROC of 0.884, outperforming training-free baselines (RAG: 0.509; Frozen DINO: 0.652), the state-of-the-art SFDA baseline SHOT (0.765), and even the supervised Oracle (0.586).

Our analysis reveals that retrieval-based approaches fail due to embedding misalignment, while supervised fine-tuning fails due to camera-artefact overfitting. Netra-Adapt succeeds by leveraging shape-based DINOv3 features and lightweight test-time calibration, offering a practical solution for deploying glaucoma screening systems across diverse populations under realistic privacy constraints.

This work establishes test-time feature calibration as a viable paradigm for cross-ethnic medical AI and opens directions for future research in single-sample adaptation and the theoretical analysis of SFDA methods.

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